WEEK-7

Bernoulli trials → only 2 possible outcomes → p = success (p), q = Failure (q)

Monte-Carlo Simulation → • Uses random sampling repeatedly
• Helps estimate probabilities when direct math is hard.

Histogram → • divides data into bins (intervals)
• shows frequency

IID Sample → • Independent
• Identically distributed $X_1, X_2, \ldots, X_n$

(empirical prob.) Key approximation → $P(A) \approx \frac{n_A}{n}$ → number of success

$n \uparrow$ accuracy $\uparrow$

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mean $= \mu$
var $= \sigma^2$
S.D $= \sigma$

$$\sigma^2 = \frac{\sum (x_i)^2}{n} - (\mu)^2$$

$$\sigma = \sqrt{\frac{\sum (x_i)^2}{n} - (\mu)^2}$$

Sample Mean → $\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$

Expectation and variance of Sample mean

$$E[\bar{X}] = \mu, \quad Var[\bar{X}] = \frac{\sigma^2}{n}$$

$$Var[\bar{x}] = \frac{Var(x)}{n} \qquad E(\bar{x}) = E(x)$$

Sample Variance → $S^2 = \frac{\sum (X_i - \bar{X})^2}{n-1}$

★★ # Expected value of Sample variance → $E[S^2] = \sigma^2$

Sample Proportion → $S(A) = \frac{\#(A \text{ occurs})}{n}$ → X_i for which A is true

$E[S(A)] = p$ → Success

$Var[S(A)] = \frac{pq}{n}$ → Failure

Expected value and Variance [sum]

Suppose $S = a_1 X_1 + \cdots + a_n X_n$, where a_i are const.

$\Rightarrow E[S] = a_1 E[X_1] + \cdots + a_n E[X_n] = (\Sigma a_i) E[X]$

$\Rightarrow Var(S) = a_1^2 Var(X_1) + \cdots + a_n^2 Var(X_n) = (\Sigma a_i^2) Var[X]$

Weak Law of Large Numbers

$$P\left(|\bar{X} - \mu| \geq \delta\right) \leq \frac{\sigma^2}{n\delta^2}$$

δ → const.

σ^2 → (const)^2

→ ≃ Chebyshev's inequality

Q. Given $n = 50$ iid i.i.d sample, Using WLLN, choose correct:

Ans. For Bernoulli (0.4) samples, with prob. more than 0.52, sample mean lies in [0.3, 0.5]

Proof: $p = 0.4$ $\quad$ ($E[x]$) mean $= 0.4 = p$

$q = 0.6$

$$\frac{\sigma^2}{n} = \frac{0.24}{50}$$

$$P\left(\bar{x} - 0.4\right) \leq 0.1 \geq 1 - \frac{\sigma^2}{n\delta^2}$$

$$\geq 1 - \frac{0.24}{50(0.1)^2}$$

$$\Rightarrow 0.52$$

Q. A random sample of size 100 is collected from a population with a S.D of 25. Find a lower bound on the probability, that the sample mean will be at most 5 units away from the actual mean using WLLN.

Sol. $$P\left(|\bar{X} - \mu| \geq \delta\right) \leq \frac{\sigma^2}{n\delta^2}$$

$$P\left(|\bar{X} - \mu| < \delta\right) \geq 1 - \frac{\sigma^2}{n\delta^2}$$

$$\geq 1 - \frac{625}{100(5)^2} \Rightarrow 0.75$$